Concept Guide: Finance & Habit Pattern Finder

Topic 9 – Google Opal

1. Introduction

Finance & Habit Pattern Finder is an AI-powered transaction analysis workflow developed using Google Opal. It uses persistent memory to maintain transaction history and analyze spending patterns across multiple workflow executions.

The project demonstrates how AI reasoning, memory tools, user inputs, and output generation can be orchestrated within a single workflow.

2. Project Objective

The primary objective is to help users understand their recorded spending activity by comparing current transactions with historical financial records.

The system is designed to:

- Capture daily transaction information.
- Retrieve previously stored transactions.
- Identify spending increases and repeated category activity.
- Detect patterns and trends from available historical data.
- Generate personalized financial recommendations.
- Append new transactions to persistent memory.
- Present insights through a browser-based financial dashboard.

3. Key Features

Transaction Input

Users provide transaction date, amount, category, and description.

Persistent Memory

The workflow retrieves historical transaction records and maintains updated transaction history across runs.

Transaction Analysis

The analysis process compares the current transaction with historical records and identifies relevant spending changes and category patterns.

Personalized Recommendations

The system generates practical suggestions based on available transaction information.

Financial Dashboard

The final output presents the latest transaction, spending insights, weekly recommendations, and transaction archive.

4. Workflow Architecture

The project follows this sequence:

1. Transaction Date – captures the transaction date.
2. Amount – captures the transaction amount.
3. Category – captures the spending category.
4. Description – captures the transaction description.
5. Process Transaction Analysis – retrieves historical memory, analyzes transactions, and updates the stored history.
6. Generate Financial Report – creates the financial insights dashboard.
7. Browser Output – displays the generated report.

5. Role of Persistent Memory

Persistent memory is a central feature of this project.

It enables the workflow to:

- Retrieve previously recorded transactions.
- Compare new transactions with historical spending.
- Preserve existing transaction records.
- Append new records without replacing old information.
- Support analysis across multiple executions.

Transaction records are maintained in a structured JSON format.

6. Example Use Case

A user records the following transactions:

Date	Description	Category	Amount
2026-11-06	Previous lunch	Food	$120
2026-11-07	Family dinner	Food	$450
2026-11-08	Restaurant lunch	Food	$180

The workflow identifies a total recorded Food expenditure of $750 across these three transactions and highlights the increase in the latest lunch cost compared with the earlier lunch.

It then generates practical recommendations for improving spending awareness.

7. Expected Benefits

- Better visibility into recorded spending.
- Awareness of repeated spending activity.
- Easier comparison of similar transactions.
- More informed personal spending decisions.
- Persistent transaction history for future analysis.

8. Scope and Limitations

This project is an educational financial analysis workflow.

It does not connect directly to bank accounts, process payments, transfer money, or provide guaranteed financial advice.

Its analysis depends on the accuracy and completeness of user inputs and available historical records.

9. Conclusion

Finance & Habit Pattern Finder demonstrates how Google Opal can combine user inputs, AI reasoning, persistent memory, and browser-based reporting to build a practical financial habit analysis workflow.